

Q1: 24 Feb (Shift 1) - Single Correct

The statement among the following that is a tautology is:

- (1) $A \wedge (A \vee B)$
- (2) $B \rightarrow [A \wedge (A \rightarrow B)]$
- (3) $A \vee (A \wedge B)$
- (4) $[A \wedge (A \rightarrow B)] \rightarrow B$

Q2: 24 Feb (Shift 2) - Single Correct

The negation of the statement

$\sim p \wedge (p \vee q)$ is :

- (1) $\sim p \wedge q$
- (2) $p \wedge \sim q$
- (3) $\sim p \vee q$
- (4) $p \vee \sim q$

Q3: 24 Feb (Shift 2) - Single Correct

For the statements p and q, consider the following compound statements:

- (a) $(\sim q \wedge (p \rightarrow q)) \rightarrow \sim p$
- (b) $((p \vee q) \wedge \sim P) \rightarrow P$

Then which of the following statements is correct?

- (1) (a) is a tautology but not (b)
- (2) (a) and (b) both are not tautologies.
- (3) (a) and (b) both are tautologies.
- (4) (b) is a tautology but not (a).

Q4: 25 Feb (Shift 1) - Single Correct

The statement $A \rightarrow (B \rightarrow A)$ is equivalent to:

(1) $A \rightarrow (A \wedge B)$

(2) $A \rightarrow (A \vee B)$

(3) $A \rightarrow (A \rightarrow B)$

(4) $A \rightarrow (A \leftrightarrow B)$

Q5: 25 Feb (Shift 2) - Single Correct

The contrapositive of the statement "If you will work, you will earn money" is:

(1) If you will not earn money, you will not work

(2) You will earn money, if you will not work

(3) If you will earn money, you will work

(4) To earn money, you need to work

Q6: 26 Feb (Shift 2) - Single Correct

Let $F_1(A, B, C) = (A \wedge \sim B) \vee [\sim C \wedge (A \vee B)] \vee \sim A$ and $F_2(A, B) = (A \vee B) \vee (B \rightarrow \sim A)$ be two logical expressions. Then :

(1) F_1 is not a tautology but F_2 is a tautology

(2) F_1 is a tautology but F_2 is not a tautology

(3) F_1 and F_2 both are tautologies

(4) Both F_1 and F_2 are not tautologies

Answer Key**Q1 (4)****Q2 (4)****Q3 (3)****Q4 (2)****Q5 (1)****Q6 (1)**

Hints and Solutions

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Q1 (4)

$$\begin{aligned} & A \wedge (\sim A \vee B) \rightarrow B \\ &= [(A \wedge \sim A) \vee (A \wedge B)] \rightarrow B \\ &= (A \wedge B) \rightarrow B \\ &= \sim A \vee \sim B \vee B \\ &= t \end{aligned}$$

Q2 (4)

p	q	$\sim p$	$p \vee q$	$(\sim p) \wedge (p \vee q)$	$\sim q$	$p \vee \sim q$
T	T	F	T	F	F	T
T	F	F	T	F	T	T
F	T	T	T	T	F	F
F	F	T	F	F	T	T

$\therefore \sim p \wedge (p \vee q) \equiv p \vee \sim q$

Q3 (3)

(a)

p	q	$\sim q$	$p \rightarrow q$	$\sim q \wedge (p \rightarrow q)$	$\sim p$	$(\sim q) \wedge (p \rightarrow q) \rightarrow \sim p$
T	T	F	T	F	F	T
T	F	T	F	F	F	T
F	T	F	T	F	T	T
F	F	T	T	T	T	T

(a) is tautologies

(b)

p	q	$p \vee q$	$\sim p$	$(p \vee q) \wedge \sim p$	$((p \vee q) \wedge \sim p) \rightarrow q$
T	T	T	F	F	T
T	F	T	F	F	T
F	T	T	T	T	T
F	F	F	T	F	T

(b) is tautologies

$\therefore$ a & b are both tautologies.

Q4 (2)

Hints and Solutions

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$$A \rightarrow (B \rightarrow A)$$

$$\Rightarrow A \rightarrow (\sim B \vee A)$$

$$\Rightarrow \sim A \vee (\sim B \vee A)$$

$$\Rightarrow \sim B \vee (\sim A \vee A)$$

$$\Rightarrow \sim B \vee t$$

= t (tautology) From options:

$$(2) A \rightarrow (A \vee B)$$

$$\Rightarrow \sim A \vee (A \vee B)$$

$$\Rightarrow (\sim A \vee A) \vee B$$

$$\Rightarrow t \vee B$$

$$\Rightarrow t$$

Q5 (1)

Contrapositive of $p \rightarrow q$ is $\sim q \rightarrow \sim p$

$p \rightarrow$ you will work

$q \rightarrow$ you will earn money

$\sim q \rightarrow$ you will not earn money

$\sim p \rightarrow$ you will not work

$\sim q \rightarrow \sim p \Rightarrow$ if you will not earn money, you will not work.

Q6 (1)

Truth table for F_1

A	B	C	$\sim A$	$\sim B$	$\sim C$	$A \vee \sim B$	$A \vee B$	$\sim C \vee (A \vee B)$	$[\sim C \wedge (A \vee B)] \vee \sim A$	$(A \wedge \sim B) \vee [\sim C \wedge (A \vee B)] \vee \sim A$
T	T	T	F	F	F	F	T	F	F	F
T	T	F	F	F	T	F	T	T	T	T
T	F	T	F	T	F	T	T	F	T	T
T	F	F	F	T	T	T	T	T	T	T
F	T	T	T	F	F	F	T	F	T	T
F	T	F	T	F	T	F	T	T	T	T
F	F	T	T	T	F	F	F	F	T	T
F	F	F	T	T	T	F	F	F	T	T

Not a tautology

Truth table for F_2

A	B	$A \vee B$	$\sim A$	$B \rightarrow \sim A$	$(A \vee B) \vee$	$(B \rightarrow \sim A)$
T	T	T	F	F	T	
T	F	T	F	T	T	
F	T	T	T	T	T	
F	F	F	T	T	T	

F_1 not shows tautology and F_2 shows tautology

Hints and Solutions

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